GEOG 4/5/7 9073: Environmental Analysis in R

Week 03.01: a quick introduction to git/GitHub and spatial data manipulation

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Today's schedule

- Open discussion
- git and GitHub
- Spatial data 101

Anything to discuss? Questions?

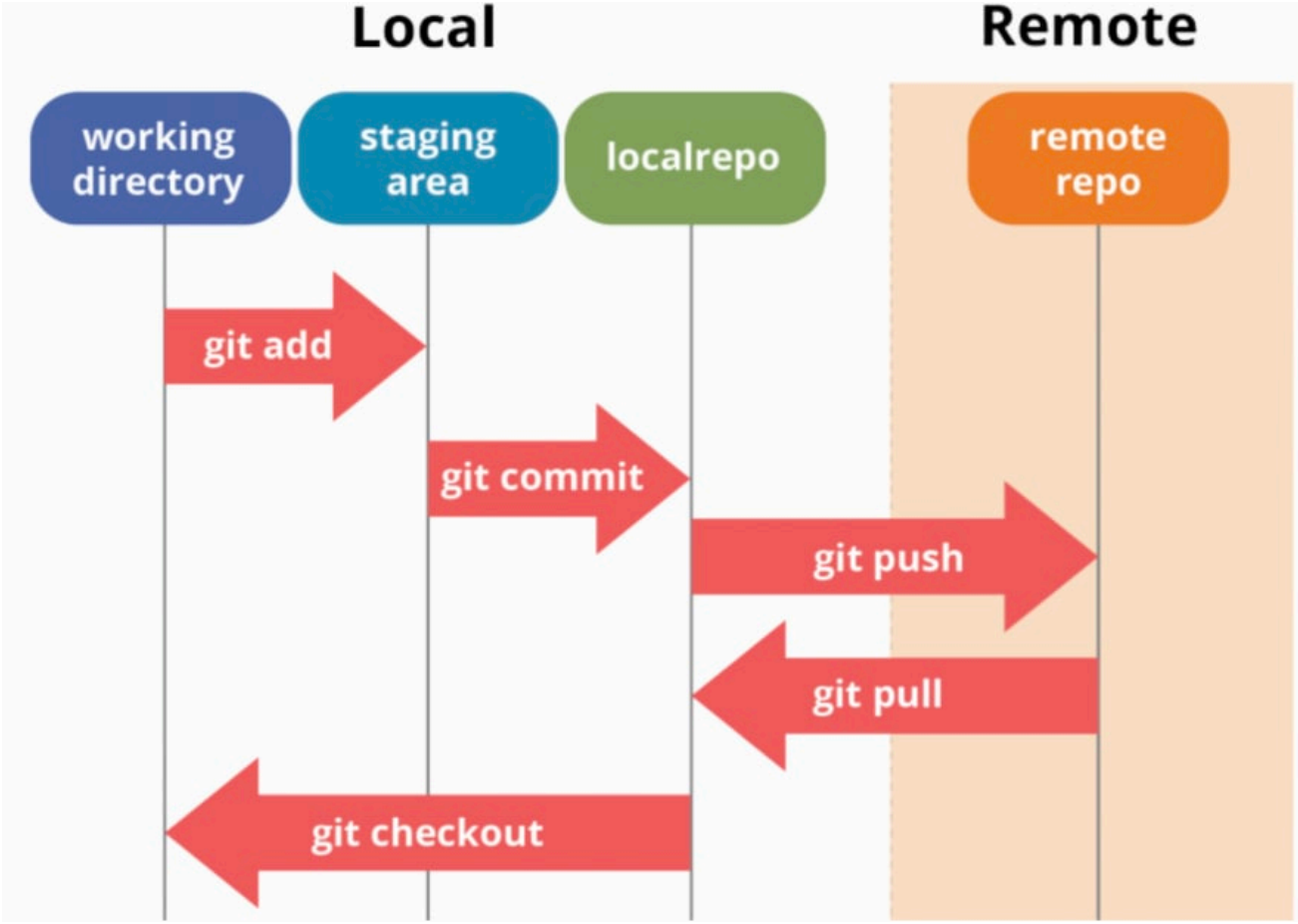
What is git and what is GitHub?

- a version control system
  - a place to store your code/data
  - a place to keep your code history
  - a place to communicate in public/private
  - a place to track/fix issues
- ...and much, much more

the "repository" (or informally, "repo")

- the "remote" repository exists on GitHub servers
- the "local" repository exists on your machine
- and you communicate between the two

The communication model



Fundamentally:

- git-commit: commit your changes locally
- git-push: push your committed changes FROM local, TO remote
- git-pull: pull the cloud version FROM remote, TO local

How you'll start the process:

1. from your GitHub account, **create** a new repo
2. from your local machine, use GitHub desktop (or other software) to first **clone** the repo locally, then
3. **pull** the repo for the first time
4. make a change (like a new RStudio project, add some data, whatever)
5. **commit** the change locally
6. then **push** to the remote (on GitHub servers)

that's it!

So when do you git-pull?

How will we use github in this course?

- it's where I'm hosting our material
- but you'll also make your own PUBLIC repository for labs (all can be in the same repo)
- and I'll use it to grade

Questions???

(I know it's a confusing topic)

A quick intro to spatial data

Today's packages

```
library(tidyverse)
library(sf)
```

New data in the course repo:

`./data/CBW/County_Boundaries.shp`

`./data/Non-Tidal_Water_Quality_Monitoring_Stations_in_the_Chesapeake_Bay.shp`

start a new project/script in R

Reading a shapefile is straightforward with *sf*

Look at it first

```
library(tidyverse)
library(sf)

p.counties <- "./data/CBW/County_Boundaries.shp"
p.stations <- "./data/CBW/Non-Tidal_Water_Quality_Monitoring_Stations_in_the_Chesapeake_Bay.shp"

d.counties <- sf::read_sf(p.counties)
d.stations <- sf::read_sf(p.stations)
```

What are we left with?

What are our initial steps?

- ESDA!
  - Investigate the objects
  - **glimpse()**
  - **plot()** <-- do NOT do this yet... you'll see why
  - But you can look at it in ArcGIS Pro or QGIS

A quick look at the data

```
glimpse(d.counties)
glimpse(d.stations)
```

Essentially a data.frame with a `geometry` attribute

- All the dplyr verbs (e.g., `select`, `filter`, `mutate`) work

So let's subset... which function would we use do that?

Let's get the counties in Delaware

Let's break it down

```
del.counties <- d.counties %>% dplyr::filter(STATEFP10 == 10)
```

A spatial problem… how do we find the stations in Delaware?

Anything we need to take into consideration prior to doing the work?

Projections 101

Our data are spatial data, so we can investigate their spatial characteristics

```
d.counties %>% sf::st_crs()
d.stations %>% sf::st_crs()
```

they're the same in this case, but we can formally test

```
d.counties %>% sf::st_crs() == d.stations %>% sf::st_crs()
```

A detour into coordinate reference systems

(or "CRS")

A CRS has 3 key elements... any ideas?

1.
2.
3.

Finding the stations in Delaware

Let's break it down

```
de.stations <- sf::st_intersection(d.stations, del.counties) # might take a bit
glimpse(de.stations)
plot(de.stations)
```

What was the output?

One last question

- does order matter?
- are these equivalent?

```
option_1 <- sf::st_intersection(d.stations, del.counties)
option_2 <- sf::st_intersection(del.counties, d.stations)
```

Yes or no?

Your tasks:

- Explore other `sf` functions (e.g., `st_area`)
- try some of what's in your book
- we'll worry about projections and mapping later

Review and next class

- Any questions?
- This week's readings/tasks:
  - Chapter 3 in textbook
  - Watch video posted to Canvas (https://www.youtube.com/watch?v=iv8rSLsi1xo)
  - Practice, practice, practice
- Lab 01 starts Thursday!